

AA Interoperability Graph APIs Specifications

v1.1

#.Vn.	Date	Author	Reviewer / Approve	Remarks
0.1	13-Oct-2026	Jithesh Kumar		Proposed AIG APIs (partial)
1.0	08-Nov-2026	Jithesh Kumar		AIG APIs, Read and Write, Applicability to AAs, FIUs, and FIPs
1.1	12-Nov-2026	Jithesh Kumar		Updated API path and token for Publish API

This is the AIG APIs and their sample payloads to be hosted on Sahamati..

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Purpose

This document describes the specifications and usage for the Sahamati's AIG APIs. AIG stands for AA-Interoperability-Graph. This graph provides information on the connectivity links that exist between entities in the AA ecosystem. The links may be between an FIP and AA, or an FIU and AA, or an AA and another AA supporting interoperability.

Entities will use this to identify routes to target entities.

Type of entity using the AIG	**Purpose**
AA	To identify a list of connected-AAs towards an FIU, and inter-AAs towards an FIP.
FIU	To identify a list of connected-AAs towards a customer-AA. In the long run, all AAs will be interconnected to each other. FIUs will not need this graph
FIP	To identify a list of inter-AAs towards an origin-AA. In the long run, all AAs will be interconnected to each other. FIPs will not need this graph

Read APIs

Lookup AIG for all FIPs

Called only by AAs

GET /v1/aig/fip

Headers:

Authorization: <Bearer token of AA>

Response 200OK:


```
{
  "ver": "1.0.0",
  "timestamp": "2026-10-07T07:35:00.000Z",
  "txnid": "1a4adbbe-29ae-11e8-a8d7-0289437bf331",
  "fip": [
   {
    "id": "fip-1",
    "inter-AA": [
     {
      "id": "aa-1",
      "verified": true,
      "attr": {
       "key1": "value1",
       "key2": "value2"
      }
     }
    ]
   }
  ]
}
```


"**verified**":

- *true*: indicates that flow through this interface has been verified (through sla data)
- *false*: indicates that interface is published, but traffic flow is yet to be verified

"attr":

- For future use; parameters indicating the health of the route

Error Response Codes:

- 400
- 401
- 403
- 429
- 500
- 501
- 503

Lookup AIG for all FIUs

Called only by AAs

GET /v1/aig/fiu

Headers:

Authorization: <Bearer token of AA>

Response:


```
{
 "ver": "1.0.0",
 "timestamp": "2026-10-07T07:35:00.000Z",
 "txnid": "3a4adbbe-29ae-11e8-a8d7-0289437bf331",
 "fiu": [
  {
   "id": "fiu-4",
   "connected-AA": [
    {
     "id": "aa-4",
     "verified": true,
     "attr": {
      "key1": "value1",
      "key2": "value2"
     }
    }
   ]
  }
 ]
}
```


"**verified**":

- *true*: indicates that flow through this interface has been verified (through sla data)
- *false*: indicates that interface is published, but traffic flow is yet to be verified

"attr":

- For future use; parameters indicating the health of the route

Error Response Codes:

- 400
- 401
- 403
- 429
- 500
- 501
- 503

Lookup AIG for all AAs

GET /v1/aig/aa

Headers:

Authorization: <Bearer token of FIU or FIP>

Response:

```
{
 "ver": "1.0.0",
 "timestamp": "2026-10-07T07:35:00.000Z",
 "txnid": "1a4adbbe-29ae-11e8-a8d7-0289437bf331",
 "aa": [
  {
   "id": "aa-1",
   "peer-AA": [
    {
     "id": "aa-2",
     "verified": true,
     "attr": {
      "key1": "value1",
      "key2": "value2"
     }
    }
   ]
  }
 ]
}
```

"**verified**":

- *true*: indicates that flow through this interface has been verified (through sla data)
- *false*: indicates that interface is published, but traffic flow is yet to be verified

"attr":

- For future use; parameters indicating the health of the route

Error Response Codes:

- 400
- 401
- 403
- 429
- 500
- 501
- 503

Write APIs

Publish AIG graph by AA

PUT /v1/aig/update/{aaid}

Headers:

Authorization: <Bearer token of User associated with the calling AA>

Request:

```
{
  "ver": "1.0.0",
  "timestamp": "2026-11-03T10:00:00.000Z",
  "txnid": "1a4adbbe-29ae-11e8-a8d7-0289437bf331",
  "fip": [
    "fip-entity-01"
  ],
  "fiu": [
    "fiu-entity-02"
  ],
  "aa": [
    "aa-entity-03"
  ]
}
```

Response 200OK:

```
{
  "ver": "1.0.0",
  "timestamp": "2026-11-06T06:22:34.258596480Z",
  "txnid": "1a4adbbe-29ae-11e8-a8d7-0289437bf331",
  "responseText": "Uploaded"
}
```

Error Response Codes:

- 400
- 401
- 403
- 429
- 500
- 501
- 503